Security Metrics and Executive Reporting:

- **MTTD**: 2 hours
  vsftpd backdoor exploitation execution
- **MTTR**: 4 hours
  CrowdSec IP block
- **FP**: 5%

150 words executive summary;

This report evaluates the response to a simulated vsftpd backdoor exploit on a Metasploitable2 instance. I achieved a Mean Time to Detect of 2 hours and a Mean Time to Respond of 4 hours. These metrics indicate that while our visibility into service-level exploits is strong, the manual transition between detection and active containment remains a bottleneck.
The False Positive rate during this attack was successfully maintained at 5%, ensuring that the focus remained on high-severity threats. To optimize future performance, it is recommended that the SOC transition from manual IP blacklisting to an automated CrowdSec integration. By automating the response to known exploit signatures, we can reduce the response window by an estimated 75%. Strengthening using these automated playbooks will significantly lower our overall risk profile and improve our defensive posture against rapid-exploitation attacks.

Dwell time analysis (50 words):

The total Dwell Time for the vsftpd exploit was 6 hours. While the 2-hour MTTD is acceptable for log ingestion, the 4-hour MTTR indicates a delay in remediation. Integrating automated playbooks to bridge the gap between Elastic alerts and CrowdSec blocks would drastically minimize this critical exposure window.